

Data Exploration

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Mean, Median, Mode

Mean

The average of the value.

```
#importing dataset
agri_dataset <- read.csv('G:/Lesson/Done/Research/R Language/R-Language/Basic/agri_dataset.csv')

#calculating mean
#The result will be NA here. Because it has some missing value.
mean(agri_dataset$Crop_Yield)
```

```
[1] NA
```

```
#here na.rm gives the command that calculate the mean without the missing value
mean(agri_dataset$Crop_Yield, na.rm= TRUE)
```

```
[1] 3476.471
```

Median

Median is the middle of the set of numbers

```
median(agri_dataset$Crop_Yield, na.rm= TRUE)
```

```
[1] 3400
```

Mode

Mode is the most frequently occurring value in the numbers

```
getmode <- function(v){  
  return(names(which.max(table(v))))  
}  
mode_of_yield <- getmode(agri_dataset$Crop_Yield)  
mode_of_yield
```

```
[1] "2500"
```

Variance and Standard deviation

Variance

```
var(agri_dataset$Crop_Yield, na.rm= TRUE)
```

```
[1] 361911.8
```

Standard deviation

```
sd(agri_dataset$Crop_Yield, na.rm= TRUE)
```

```
[1] 601.591
```

Quartile

Provide insights into the distribution of the data

```
dataset_q <- c(1,2,3,4,5,6,7,8,9,10,11,12,13,14,15)
quantile(dataset_q,probs = c(0.25,0.5,0.75))
```

```
25%  50%  75%
4.5   8.0 11.5
```

```
quantile(agri_dataset$Crop_Yield,probs=c(0.25,0.5,0.75),na.rm=TRUE)
```

```
25%  50%  75%
3000 3400 4000
```